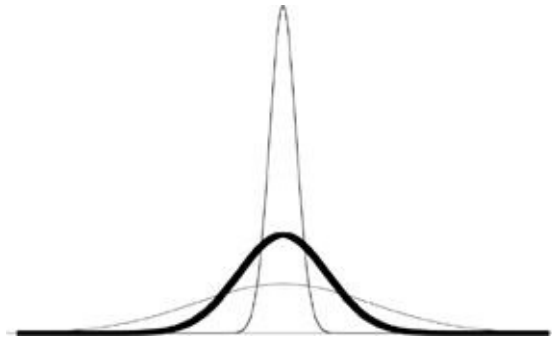




In Mediocristan, as your sample size increases, the observed average will present itself with less and less dispersion—as you can see, the distribution will be narrower and narrower. This, in a nutshell, is how everything in statistical theory works (or is supposed to work). Uncertainty in Mediocristan vanishes under averaging. This illustrates the hackneyed “law of large numbers.”

The safety of my coffee cup illustrates how the randomness of the Gaussian is tamable by averaging. If my cup were one large particle, or acted as one, then its jumping would be a problem. But my cup is the sum of trillions of very small particles.

Casino operators understand this well, which is why they never (if they do things right) lose money. They simply do not let one gambler make a massive bet, instead preferring to have plenty of gamblers make series of bets of limited size. Gamblers may bet a total of \$20 million, but you needn’t worry about the casino’s health: the bets run, say, \$20 on average; the casino caps the bets at a maximum that will allow the casino owners to sleep at night. So the variations in the casino’s returns are going to be ridiculously small, no matter the total gambling activity. You will not see anyone leaving the casino with \$1 billion—in the lifetime of this universe.

The above is an application of the supreme law of Mediocristan: when you have plenty of gamblers, no single gambler will impact the total more than minutely.

The consequence of this is that variations around the average of the Gaussian, also called “errors,” are not truly worrisome. They are small and they wash out. They are domesticated fluctuations around the mean.

Love of Certainties

If you ever took a (dull) statistics class in college, did not understand much of what the professor was excited about, and wondered what “standard deviation” meant, there is nothing to worry about. The notion of standard deviation is meaningless outside of Mediocristan. Clearly it would have been more beneficial, and certainly more entertaining, to have taken classes in the neurobiology of aesthetics or postcolonial African dance, and this is easy to see empirically.